

Project 6

Offset-Wrist IK through the FANUC CRX Case Study

Format: Group of two students; simulation-based project.

Prerequisites: Rigid transforms, D–H modelling, numerical inverse kinematics, Jacobians, singularities, and polynomial root finding.

Motivation

The FANUC CRX provides a concrete example of an offset-wrist 6R robot whose inverse kinematics cannot be treated as the PUMA arm–spherical-wrist decomposition. Studying a richer solution set reveals why IK enumeration and global connectivity must be examined separately before planning a motion.

Task

Overall objectives. Use the supplied CRX-10iA/L IK model to study solution multiplicity, singularities, and selected connections between IK solutions. Federico Thomas and Josep M. Porta have developed the robot’s closure-polynomial formulation; the numerical IK implementation will be provided, so the project focuses on interpreting its solutions.

- **Validate the CRX model.** Inspect the supplied URDF/STL geometry and document the D–H model, joint signs and offsets, and base/tool frames used by the IK implementation. Compare forward kinematics from both representations. Explain geometrically why a spherical-wrist decomposition is unavailable.
- **Understand the polynomial formulation.** Read the CRX case in *The inverse kinematics of lobster arms*. Describe the distance variable, closure polynomial, and recovery of joint configurations. Distinguish an explicit polynomial formulation from a formula for its roots in radicals; re-deriving the full polynomial is not required.
- **Inspect and validate solution multiplicity.** Reproduce the supplied sixteen-solution reference case after checking the angle convention. Verify position and orientation closure for every solution, deduplicate periodic equivalents, and report joint-limit admissibility separately. Compare two nearby target poses using the supplied solver.
- **Visualize singularities and IK locations.** Evaluate $\det J$ on two selected two-joint slices, stating all fixed joint values; choose a (q_2, q_3) slice for at least one plot. A projected IK whose other joints differ does not belong to that slice: distinguish projections from on-slice points. Supplement the plots with full-Jacobian singular values at every IK.
- **Investigate one connectivity question.** Use an instructor-supplied candidate joint path between two IKs of a common pose. Check endpoint pose agreement, full-Jacobian regularity, and the closed workspace image of the path. Explain why the existence of sixteen solutions or equal determinant signs alone does not establish their connectivity or cuspidality.

Specifics and scope

Use one reference target, two small pose perturbations, two joint-space slices, and one supplied candidate path. The IK model and candidate path will be provided. Numerical polynomial solution, full-pose checks, and interpretation are required; deriving the general 6R IK polynomial or searching the entire six-dimensional configuration space is outside scope. State the scaling used for the mixed linear/angular Jacobian.

Check both translation and orientation residuals for every claimed full-pose IK. Document angle periodicity and limits, Jacobian conventions, and any scaling used for singular values. State all fixed coordinates in reduced plots; a two-dimensional slice does not show the entire 6R singular set. Refine decisive transitions, and distinguish sampled evidence from a continuous-path guarantee. Collision freedom is not implied by these kinematic checks.

Expected outcome

Understand CRX geometry, the role of closure polynomials, the richer IK distribution, and the distinction between solution count and connectivity. Deliver a validated IK table, singularity slices, and an evidence-based analysis of the candidate path and its workspace image.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Provided materials and implementation

The CRX-10iA/L robot description (.urdf) and link meshes (.stl) will be provided to the students, together with the numerical IK model, sixteen-solution reference data, and candidate joint path. Use the same model variant for all comparisons. The reference joint angles are given in degrees in a D–H convention. Let θ_i denote these angles converted to radians, and map them to the supplied URDF convention using

$$q_i = \text{wrap}_{[-\pi, \pi)}(s_i \theta_i + b_i), \\ s = (1, -1, -1, 1, -1, -1), \quad b = (0, -\pi/2, 0, \pi, \pi, 0).$$

Check this conversion by comparing all reference forward poses. Wrapped angles are representatives modulo 2π ; check physical limits separately and keep joint trajectories continuous when validating the candidate path.

Suggested reading: Federico Thomas and Josep M. Porta, *The inverse kinematics of lobster arms*, Mechanism and Machine Theory 196 (2024), 105630. The paper develops a distance-based closure-polynomial formulation, including the CRX-10iA/L case.

Use a robotics library or your own tools to verify D–H parameters, inspect axes and frames, visualize the recovered IKs and singularity slices, and animate the candidate path. Validate numerical results independently with forward kinematics and the full Jacobian.